# Long-Term Changes of Chlorophyll-a in Lake Okeechobee: Combining the Strengths of In Situ Observations, Multi-Sensor Remote Sensing, and Machine Learning

Cheng Xue, Jennifer P. Cannizzaro, Chuanmin Hu, Brian B. Barnes, Yuyuan Xie, Lin Qi, Cassondra Armstrong, Zhiqiang Chen, and Paul R. Jones

***Abstract*— Many lakes around the world have faced eutrophication and frequent cyanobacterial harmful algal blooms (cyanoHABs), where chlorophyll-a (Chla) concentration is often used to determine the trophic status of the lake and evaluate the effectiveness of nutrient reduction strategies. However, limited in situ sampling and the absence of robust remotely sensed data product often hinder the analysis of long-term Chla changes in such lakes. Here, using Lake Okeechobee (LO) as an example, we addressed this challenge by developing a novel method to harnesses the power of both multi-sensor remote sensing and machine learning (ML). The method first used in situ Chla to estimate Chla from OLCI imagery ($Chla_{OLCI}$), which then served as a surrogate for in situ data toward obtaining the large matchup datasets for MODIS model training. Subsequently, two deep neural network (DNN) models, $DNN_S$ and $DNN_{NS}$, were trained for the MODIS-saturated pixels with high algal biomass waters and the non-saturated pixels with lower Chla but more turbid waters, respectively. The daily and monthly Chla derived from MODIS ($Chla_{MODIS}$) were validated using in situ Chla ($R^2 \sim 0.77$, Mean absolute percentage difference (MAPD) ~25%) and monthly $Chla_{OLCI}$ ($R^2 \sim 0.90$, MAPD ~ 11%), respectively. Monthly $Chla_{MODIS}$ derived from Terra in 2000—2023 was used to analyze the spatiotemporal changes in LO. We observed a significant long-term increasing trend of Chla (~2 mg $m^{-3}$ $decade^{-1}$) and a ~30% increase relative to the multiple-year average. To explore the potential factors driving long-term Chla changes, water temperature, precipitation, inflow, stage, and the multivariate ENSO index (MEI) were tested through multi-variative analysis. Although further testing is required, the approach of using one sensor as a "bridge" to bring limited in situ data and another sensor together to overcome the data scarcity and lack of spectral bands is believed to be extendable to other eutrophic lakes around the world.**

***Index Terms*— Chlorophyll-a, cyanobacterial bloom, Lake Okeechobee (LO), long-term trend, machine learning (ML), remote sensing.**

## I. Introduction

CHANGES in lake dynamics can have broad impacts on aquatic ecosystems and anthropogenic activities, while eutrophication in lake waters worldwide is one of the greatest environmental problems facing humanity [1], [2]. As the second largest lake located entirely in the contiguous USA, Lake Okeechobee (LO) serves as the central component of the hydrology and environment of South Florida, acting as a reservoir for potable and irrigation water for much of south Florida, and supporting valuable commercial and sport fisheries [3], [4], [5]. However, LO has been subject to several long-term stresses, including a substantial water warming rate, excessive nutrient loads, and extreme water level fluctuations, which have been linked to cyanobacterial harmful algal blooms (cyanoHABs) [6], [7], [8], [9]. CyanoHABs primarily occur in summer and can pose a risk to public health, especially when bloom-laden water is released to more highly populated downstream estuaries [10], [11].

Long-term, massive efforts have been established in the watershed to mitigate eutrophication impacts. This includes establishing a legally mandated target of total phosphorus (TPs) loads to LO [12], [13], but annual P-loads that exceed the target value have persisted in recent decades [14]. Chlorophyll-a (Chla) concentration, a key proxy for algal biomass, was selected to serve as the main index for determining the effectiveness of management actions to reduced nutrient loading in LO [15], [16], [17], [18], [19], [20]. Based on in situ Chla collected regularly for water quality monitoring, the frequency of bloom occurrence is determined as the percent of samples with Chla > 40 mg/$m^3$ [21]. However, in situ Chla is inadequate for analyzing the long-term changes of Chla in the entire LO because of the limitations of the sites and sampling frequency [5], [22]. Moreover, in situ Chla data are also inadequate when making reservoir release decisions

Received 22 December 2024; revised 20 February 2025 and 22 March 2025; accepted 23 April 2025. Date of publication 6 May 2025; date of current version 22 May 2025. This work was supported in part by the U.S. National Aeronautics and Space Administration (NASA) Water Resources Program under Grant 80NSSC19K1200; and in part by the Ocean Biology and Biogeochemistry Program under Grant 80NSSC22K1299, Grant 80NSSC20M0204, and Grant 80NSSC24K1417. *(Corresponding author: Chuanmin Hu.)*

Cheng Xue, Jennifer P. Cannizzaro, Chuanmin Hu, Brian B. Barnes, and Yuyuan Xie are with the College of Marine Science, University of South Florida, St. Petersburg, FL 33701 USA (e-mail: chengxue@usf.edu; jpatch@usf.edu; huc@usf.edu; bbarnes4@usf.edu; yuyuan@usf.edu).

Lin Qi is with the NOAA Center for Satellite Applications and Research, College Park, MD 20740 USA, and also with Global Science and Technology Inc., Greenbelt, MD 20770 USA (e-mail: lin.qi@noaa.gov).

Cassondra Armstrong, Zhiqiang Chen, and Paul R. Jones are with South Florida Water Management District, West Palm Beach, FL 33406 USA (e-mail: cthomas@sfwmd.gov; zhichen@sfwmd.gov; pajones@sfwmd.gov).

Digital Object Identifier 10.1109/TGRS.2025.3567318

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

aimed at preventing the transport of toxic cyanoHABs downstream to highly populated areas adjacent to estuaries [23]. Thus, it is necessary to increase the frequency and coverage for Chla observations in LO to determine the long-term trend of Chla and further assist the water management agencies in decision making [5].

Remote sensing offers an improvement over field sampling by providing synoptic coverage in both space and time. It has been used in monitoring long-term Chla changes in many inland waters using multiple sensors with variable spectral, spatial, and temporal resolution and lifespans [24], [25], [26], [27], [28], [29], [30]. Nevertheless, no promising long-term data product of remotely sensed Chla is currently available for LO due to optical complexity, including high and variable Chla and turbidity [31], [32]. This is despite the progress made toward the use of the Cyanobacteria Index (CI) [33] as an indicator of cyanoHABs using 300 m-resolution data obtained from MERIS aboard ENVISAT (2002–2012) and OLCI aboard Sentinel-3A (2016-) and Sentinel-3B (2018-) [23], [34], [35], [36]. The index provides a strong indicator of blooms (Chla $>$ 40 mg $m^{-3}$), with waters having lower Chla being classified as non-bloom. Application of existing remote sensing algorithms did not lead to meaningful results for these waters (see Section II-E). As a result, no quantitative, accurate remotely sensed Cha time series has been attained in LO, thus hindering any analysis of long-term changes in Chla.

The goal of this study is to fill this knowledge gap through development of a validated remote sensing algorithm and derivation of long-term Chla patterns in LO, with the algorithm being applicable for a wide dynamic range. Specifically, we have the following objectives: 1) determine a hybrid model for OLCI applicable to optically complex bloom and turbid, non-bloom waters in LO based directly on in situ Chla; 2) develop machine learning (ML) models for MODIS (fewer spectral bands and limited radiometric range, but longer time series than OLCI) to indirectly obtain Chla using OLCI Chla as a surrogate; and 3) examine spatiotemporal Chla variations and long-term Chla trends using the MODIS data record (2000–2023).

The knowledge of long-term changes of Chla in LO is the first step toward the understanding of the factors contributing to such changes, which will aid in improved decision-making for the water management efforts in LO. This study combines the strengths of in situ observations, multi-sensor remote sensing, and ML to improve our knowledge on Chla changes. Because LO is not the only lake facing such a technical challenge, the approach developed here is expected to be applicable to other similar water bodies.

## II. Data and Methods

### A. Study Area

LO is a highly modified and managed subtropical lake in the Kissimmee–Okeechobee–Everglades watershed, consisting of 78% open water and 22% littoral habitat [37] (see Fig. 1). Although LO is large (1730 $km^2$) in areal extent, it is relatively shallow with a mean depth of ~2.7 m. The climate and rainfall in the area is characteristic of subtropical locations

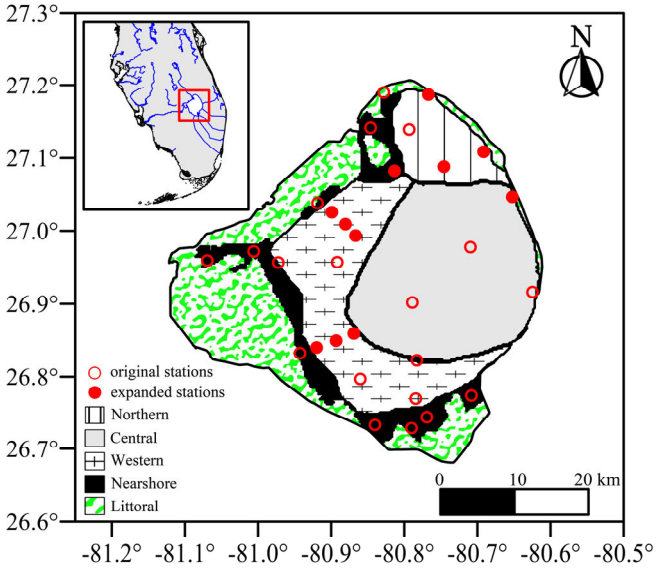



Fig. 1. Map of the LO study area located in south Florida (USA) with water quality sampling stations, four major ecological zones [37], and the littoral zone. The shallow (<2 m) littoral and nearshore zones were excluded from the following data analysis. The filled red dots show the expanded water quality sampling stations after 2020, while the red unfilled circles show the 19 original stations (see Section II-B).

with a dry (November–April) and wet (May–October) season. Major inflows of water from several creeks and rivers located on the northern and western sides of the lake together with direct rainfall are the fundamental input sources of water (~50% each). The outputs include evapotranspiration (~67%) and controlled water releases (~33%) both west and east to downstream estuaries and south to the Everglades Agricultural Area.

There are four major ecological zones (Central, Northern, Western, and Nearshore) that encompass the open water and nearshore portions of the lake [37] (see Fig. 1). These zones exhibit spatial heterogeneity about trophic state parameters and relationships between phytoplankton standing crop and environmental factors [19], [37]. Due to the deeper water depth (~4–5 m), phytoplankton standing stock (as measured by Chla) is relatively low in the Central zone throughout the year, as algal growth is primarily controlled by light limitation caused by frequent wind resuspension of highly flocculent muddy sediment. Phytoplankton growth in the Northern zone (~2–4 m deep), which receives the highest volume of nutrient-laden freshwater inputs, is generally controlled by light limitation in winter and nitrogen limitation in summer/fall. The Western zone (~2–4 m) serves as a transition between the deeper, turbid Central zone and shallower (~1–2 m) Nearshore zone. Phytoplankton standing stock in the Western zone is generally high because of sufficient nutrient loads and reduced light limitation. Sand and rock/peat bottoms make this region less vulnerable to wind-driven sediment resuspension. The Nearshore zone, located along the western and southern sides of LO, can support submerged aquatic vegetation, though coverage is variable from year to year. Given the shallow depths (<2 m) and increased likelihood for emergent/submerged aquatic vegetation that may interfere with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

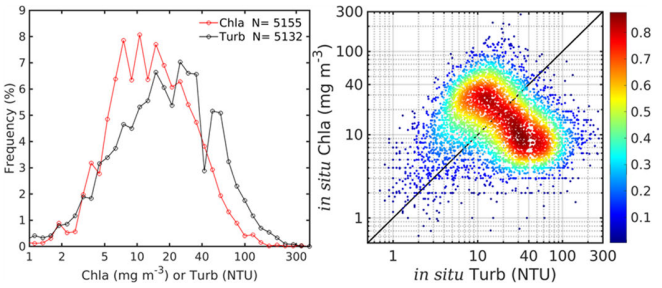



Fig. 2. Distribution of in situ Chla and Turb between 2002 and 2023 in the entire LO. The color in the scatter plot represents the probability density of the points.

satellite reflectance measurements, the littoral and nearshore zones were excluded from the following data processing and analysis.

### B. In Situ Water Quality Data

Water quality of LO has been routinely measured by the South Florida Water Management District (SFWMD), and rigorous quality assurance procedures have been implemented. Fig. 1 shows the location of the water quality sampling stations across the entire lake. Near-surface water samples (0.5 m below the water surface) were collected monthly at 19 sites between 2002 and 2020. From 2020 to 2023, the total number of sites visited expanded to 30 and the sampling frequency during the wet season was increased to bimonthly. The in situ Chla (in mg $m^{-3}$) was used to develop the remote sensing model, and both in situ Chla and Turb (in NTU) were used to indicate the optical complexity of LO. The water quality data can be accessed through the SFWMD's environmental database (DBHYDRO, https://www.sfwmd.gov/science-data/dbhydro).

Fig. 2(a) shows the histograms of all in situ Chla and Turb, and Fig. 2(b) shows the corresponding scatterplot, indicating a lack of correlation between these parameters. Chla and Turb ranged from 1 to 300 mg $m^{-3}$ and 1–300 NTU with histogram peaks at about 10 mg $m^{-3}$ and 20 NTU, respectively. The percentage of in situ data showing Chla less than the bloom threshold (Chla $<$ 40 mg $m^{-3}$) and high turbidity (Turb $>$ 10 NTU) is ~63%, indicating the dominance of turbid, non-bloom waters in LO and highlighting the need for a remote sensing approach that is capable of resolving Chla in such waters.

### C. Other Environmental Data

Several environmental parameters were tested using multiple linear regression (the ordinary least squares method in the statsmodels package of Python) to explore the potential driving factors responsible for temporal changes in Chla [38]. Monthly water surface temperature data were obtained from the MODIS daytime sea surface temperature (SST) data product, which was validated using in situ data collected from LO [7]. Monthly precipitation (rainfall) data for the LO rain area was obtained using Next Generation Radar (NEXRAD) technology from SFWMD (https://apps.sfwmd.gov/nexrad2/nrdmain.action), and was used to indicate when large pulses of nutrient-laden fresh water was added to the system. Rainfall may also serve as a proxy for extreme weather events (or tropical storms) when high winds often have a significant impact on turbidity. Monthly inflow data provided by SFWMD was used as the surrogate for nutrient loads given the high correlation between inflow volume and nutrient loads [39]. High water level is typically correlated with bloom activity in LO, likely due to a combination of internal nutrient transport and high nutrient inflows typical of wetter years. As such, lake stage data were downloaded from DBHYDRO database (station: LAKEOKEE) and included in the analyses. The Multivariate ENSO Index (MEI) (https://psl.noaa.gov/enso/mei/) was also tested because higher than normal precipitation and discharge typically occurs in South Florida during the dry season of El Niño years [40].

### D. Satellite Data

MODIS onboard Terra (2000-) and Aqua (2002-) and Sentinel-3A/B (hereafter S3A/B) OLCI data were used in this study. MODIS is suitable for trend analysis because it is capable of providing $>$20 years of multi-band optical measurements. OLCI was used despite its shorter lifespan because of its higher spatial resolution (300-m) and increased number of spectral bands compared to MODIS. Also, it boasts a higher radiometric range, thus avoiding the MODIS red-NIR band saturation issue over bright targets (e.g., intense algal blooms) [41].

Level-0 data of MODIS/Terra (February 2000–December 2023) and MODIS/Aqua (July 2002–December 2023), and the full resolution (300 m) Level-1B data of OLCI/S3A (April 2016–December 2023) and OLCI/S3B (May 2018–December 2023) were downloaded from NASA OB.DAAC (https://oceancolor.gsfc.nasa.gov). All data were processed to Level-2 data using the OCSSW module (SeaDAS version 8.3, NASA Reprocessing 2022.0) to obtain spectral Rayleigh-corrected reflectance ($R_{rc}(\lambda)$, dimensionless) [42] and CI; dimensionless) [33], [41]. The reason for using partially corrected $R_{rc}(\lambda)$ instead of the fully atmospherically corrected spectral remote sensing reflectance ($R_{rs}(\lambda)$; units: $sr^{-1}$) as the input for all models is because standard atmospheric correction approaches based on the "black pixel" assumption fail in optically complex inland waters [43], [44]. The OCSSW configurations for processing OLCI data followed those for OBPG Cyanobacteria Assessment Network (CyAN) version 5 data because of the improvements of this version in highly turbid waters [45]. Then, Level-2 data were reprojected into 250-m Level-3 data. For data quality control, the flags of LAND, HIGLINT, and HISATZEN in l2_flags [46] and the habs_flags were used. The pixels associated with cloud, land adjacency, and/or other non-water features were masked out with non-zero habs_flags. The habs_flags of MODIS data were identified using land bands, so the pixels where ocean bands including 667, 678, and 748 nm saturated due to cloud were further excluded from MODIS data using these ocean bands. The traditionally used HILT and CLDICE flags in l2_flags was not applied here since this study utilizes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

$R_{rc}$ rather than $R_{rs}$ and habs_flags could redefine the pixels flagged with CLDICE.

CI in its original form was formulated as a spectral shape of the form of a second derivative centered about 681 nm [33], which is based on the bands of 665, 681, and 709 nm on OLCI (denoted as $CI_{OLCI}$). $CI_{cyano}$, which takes into consideration a second spectral derivative with 620 nm and is used to discriminate between cyanobacterial and non-cyanobacterial blooms, was not applied in this study because LO blooms are mainly cyano-dominated [47] and the objective of deriving Chla in the lake focuses on total algal biomass regardless of the bloom type. To obtain CI for MODIS (denoted as $CI_{MODIS}$), the ocean bands 667, 678, and 748 nm were adopted [41]. Although MODIS band placement is less ideal because it lacks a 709 nm band that is more sensitive to Chla and the radiometric range is limited, strong agreement between $CI_{MODIS}$ and $CI_{OLCI}$ was observed when a correction factor of 2.71 was applied [48]. The equations of $CI_{OLCI}$ and $CI_{MODIS}$ are as follows, respectively,

$$CI_{OLCI} = (R_{rc}(709) - R_{rc}(665)) \times \frac{681 - 665}{709 - 665} - (R_{rc}(681) - R_{rc}(665)) \quad (1)$$

$$CI_{MODIS} = 2.71 \times \left( (R_{rc}(748) - R_{rc}(667)) \times \frac{678 - 667}{748 - 667} - (R_{rc}(678) - R_{rc}(667)) \right). \quad (2)$$

### E. Development of Chla Inversion Method

*1) Practical Considerations:* As mentioned earlier, the CI model [33], [41] provides a meaningful bloom index for hypereutrophic lakes, particularly for bloom waters with CI > 0 where CI highly correlates with Chla. Tomlinson et al. [49] showed strong linearity between in situ Chla (> 20 mg $m^{-3}$) and CI for several Florida lakes. However, this index does not work for turbid, low-Chla waters where CI ≤ 0 (equivalent to Chla < ∼20 mg $m^{-3}$) [23], [49]. Therefore, alternative approaches must be developed to cover a wider dynamic range of Chla. Furthermore, the OLCI data record currently goes back only ∼ seven years, thus the use of the longer data record from MODIS/Terra (2000–present) and MODIS/Aqua (2002–present) is desirable. However, MODIS has fewer spectral bands than OLCI (especially with the lack of the 709-nm band), and the spatial resolution (1 km) is also lower than OLCI (300 m). The saturation of the MODIS 678-nm band over intense blooms is another problem [41]. Clearly, in order to derive long-term Chla patterns using MODIS to cover a full dynamic range of Chla, some novel methods need to be developed.

To overcome such challenges, we first attempted ML models using matching pairs between MODIS $R_{rc}(\lambda)$ observations and in situ Chla data. These ML models included extreme gradient boost (XGB), random forest (RF), support vector machine (SVM), and neural network (NN), which have shown success in other studies [25], [44], [50], [51], [52]. NN models typically work well for large datasets (thousands of data pairs) but XGB, RF, and SVM models are more suitable for smaller datasets (hundreds of data pairs) [30], [53]. However, the results from our first attempt were not satisfactory. After many trials and errors, the best model yielded $R^2$ of 0.6, which could not be further improved even when the model training reached the point of overfitting. Combined with the aforesaid fewer bands and lower resolution for MODIS, we also attributed this non-satisfactory performance to the limited matching pairs ($N < 400$) and even fewer matching pairs ($N < 20$) for saturated MODIS pixels (i.e., high Chla pixels). Therefore, it is difficult to establish a reliable inversion method for MODIS based directly on in situ Chla to study spatiotemporal patterns and long-term changes of Chla.

Deep-learning (DL) models, an advanced form of ML that utilizes NN with multiple layers, have been used in remote sensing to extract image features and estimate water quality parameters [24], [52], [54], [55], [56], [57], [58]. DL models can learn spectral features directly from the data without the need for manual selection and combination of spectral bands because of the ability to automatically extract and learn hierarchical features through multiple layers of nonlinear transformations, which is particularly useful for optically complex waters where spectral information is rich and intricate [58], [59]. However, a critical requirement of DL models is the large data quantity for model training [59], which cannot be met by using MODIS and in situ data directly [58], [60].

For these reasons, this study uses OLCI as a bridge to increase the number of data pairs in the DL model training, such that improved model performance can be achieved. In this "indirect" method, a model (see Section II-E2 below) to estimate Chla from OLCI imagery (hereafter $Chla_{OLCI}$) was first developed by using matching in situ data. Then, $Chla_{OLCI}$ was used as a surrogate as in situ data (i.e., "ground truth") to train a Chla model used for MODIS ($Chla_{MODIS}$). Because the quantity of matching pairs between OLCI and MODIS observations are nearly infinite for both non-saturated and saturated pixels, this facilitated the development of a deep NN (DNN), which refers to the NN with more than three hidden layers) that was trained separately for non-saturated ($DNN_{NS}$) and saturated ($DNN_S$) pixels [54], [61].

Fig. 3 shows the flowchart of the proposed "indirect" method, where a hybrid model was first established to derive $Chla_{OLCI}$ (see Section II-E2), which was then used to estimate $Chla_{MODIS}$ (Section II-E3). The coefficient of determination ($R^2$), root mean squared difference (RMSD), and mean absolute percentage difference (MAPD) were calculated to assess the model performance.

*2) ChlaOLCI, an Intermediate Product Toward $Chla_{MODIS}$:* A hybrid model [see (3)] was used to derive $Chla_{OLCI}$, which estimated Chla using a CI model [$Chla_{OLCI}$(CI)] [33] and an XGB model [$Chla_{OLCI}$(XGB)] for $CI_{OLCI} > 0$ and $CI_{OLCI} \leq 0$, respectively,

$$Chla_{OLCI} = \begin{cases} Chla_{OLCI}(CI), & CI_{OLCI} > 0 \\ Chla_{OLCI}(XGB), & CI_{OLCI} \leq 0. \end{cases} \quad (3)$$

This hybrid method is similar to approaches whereby different algorithm formulations or models are selected based on Optical Water Type, which have been successful for determining Chla in other optically complex inland waters [62], [63].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

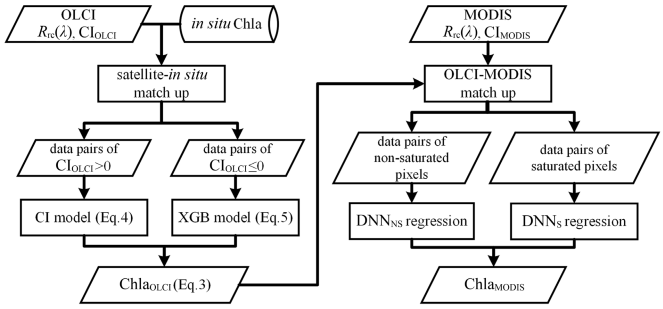



Fig. 3. Workflow for deriving $Chla_{MODIS}$ through the intermediate $Chla_{OLCI}$. The reason for using the intermediate $Chla_{OLCI}$ as a "bridge" is because the model using direct in situ Chla in the model training was not satisfactory.

*a) Matchups between in situ and OLCI data:* To maximize the number of matched data pairs, OLCI/S3A/B and in situ Chla data observed on the same day in 2016–2023 were screened. $R_{rc}$ and $CI_{OLCI}$ data were taken from a 3 × 3-pixel window centered near the in situ sampling locations. The pixels within 2 km from the lake shoreline were excluded to avoid land adjacency effects [61]. The coefficient of variation (CV) was calculated as the standard deviation divided by the mean within the window. The pixels with the CV average of $R_{rc}(665)$, $R_{rc}(681)$, and $R_{rc}(709)$ greater than 15% were excluded from the window. The averages of spectral $R_{rc}$ and $CI_{OLCI}$ were obtained from the windows with nine valid pixels.

When establishing the CI model (see below), some outliers in the matched data pairs were excluded to minimize the impact of spatial and temporal variability between the in situ Chla and satellite observations [64], [65]. Specifically, the following match-up pairs were excluded: 1) in situ Chla > 90 mg $m^{-3}$ with $CI_{OLCI} < 0.01$ and 2) in situ Chla > 40 mg $m^{-3}$ with $CI < 0$. Consequently, 56 match-up pairs with $CI_{OLCI} > 0$ were identified for developing the CI model. For the pixels of $CI_{OLCI} \leq 0$ used to develop the XGB model, additional visual examination of the imagery was performed to release some overmasked pixels with high turbidity and omit some undermasked pixels with hazy atmospheric conditions, as well as excluding matchups with Chla < 3 mg $m^{-3}$ because of too few numbers. Thus, 269 data pairs were ultimately used for developing the XGB model.

*b) CI model:* The CI model [33] was the empirical linear relationship ($R^2 = 0.62$, $N = 325$) between in situ Chla and $CI_{OLCI}$ [see (1)] that was applied to the pixels with $CI_{OLCI} > 0$

$$Chla_{OLCI}(CI) = 9689 \times CI_{OLCI} + 21.93, CI_{OLCI} > 0. \quad (4)$$

*c) XGB model:* The ML model was determined using the OLCI $R_{rc}(\lambda)$-in situ Chla match-up pairs where $CI_{OLCI} \leq 0$. These matchups were further randomly separated into training (70%, $N = 189$) and testing (30%, $N = 80$) datasets. XGB, RF, and SVM models were all tested with the same data pairs and following the same process, including the input feature selection, cross validation, and hyperparameter tuning. Of them, the XGB model outperformed on the testing set, so the following description is focused on XGB. The input features data were standardized using the mean and standard deviation within the 0–1 range, and the output data (i.e., in situ Chla) was log-transformed. The XGB regression model was implemented using the Python API (https://xgboost.readthedocs.io/en/stable/python/python_api.html) with tuning hyperparameters, such as the learning rate, the number of trees, the tree depth, and the number of samples [66]. To avoid the problem of overfitting and search for the optimal hyperparameters, the parameterization of XGB was based on the training set using Grid Search (https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html). Cross-validation was used to prevent overfitting in model training, which was conducted in the GridSeachCV function with a default fivefold cross validation. With Grid Search, the predefined ranges and increments of hyperparameters were used to fit, and then the performances with all combinations of hyperparameter values were evaluated. The set of hyperparameter values leading to the most accurate estimation results were finally used to retain the XGB model for pixels of $CI_{OLCI} \leq 0$ and generate $Chla_{OLCI}$(XGB).

$R_{rc}(\lambda)$ and associated combinations served as the input features based on the previous studies [25], [29], [44], [48], [67]. Following the parameterization process, those combinations were tested and screened based on the performance on the testing set. The tested combinations included all $R_{rc}$ from individual bands, the $R_{rc}$ ratios of two bands, the spectral slopes between two adjacent bands, the 3- and 4-band combinations of bands > 600 nm including $(R_{rc}(\lambda_1)^{-1} - R_{rc}(\lambda_2)^{-1}) \times R_{rc}(\lambda_3)$ and $(R_{rc}(\lambda_1)^{-1} - R_{rc}(\lambda_2)^{-1}) \times (R_{rc}(\lambda_3)^{-1} - R_{rc}(\lambda_4)^{-1})$, and some published spectral indexes, such as CI and the Baseline Normalized Difference Bloom Index (BNDBI) [29], [44], [67]. Finally, the following input features were used: BNDBI (calculated from 490, 560, 620, and 865 nm), $R_{rc}(412)/R_{rc}(865)$, $R_{rc}(754)/R_{rc}(865)$, $R_{rc}(674)/R_{rc}(560)$, $R_{rc}(674)/R_{rc}(709)$, $(R_{rc}(674)^{-1} - R_{rc}(709)^{-1}) \times R_{rc}(754)$, $R_{rc}(560)$, and $R_{rc}(709)$. And the corresponding optimal hyperparameters included learning_rate = 0.04, n_estimators = 130, max_depth = 4, and subsample = 0.5. Additionally, the output of XGB model, $Chla_{XGB}$, was further adjusted using (5) to force the scatter comparison close to the 1:1 line, resulting in the fine-tuned XGB output, i.e., the $Chla_{OLCI}$(XGB)

$$Chla_{OLCI}(XGB) = 1.2269 \times Chla_{XGB} - 1.8867, CI_{OLCI} \leq 0. \quad (5)$$

*3) $Chla_{MODIS}$, the Final Product to Study Spatial Patterns and Long-Term Changes:*

*a) Matchups between OLCI and MODIS data:* OLCI/S3A and MODIS (including Terra and Aqua to maximize the number of matched data pairs) data collected on the same day in 2016–2023 were screened to generate a large and robust match-up dataset for training the DNN models. Prior to generating these match-ups, a coarse cloud mask was manually drawn and used for OLCI/S3A to avoid the effects of thin clouds or hazy weather. The images with sensor zenith angle greater than 50° were excluded. The pixels within 2 km from the lake shoreline were excluded. The spatial homogeneity at the co-located pixel was tested using CV within a 3 ×

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

TABLE I
INPUT FEATURES AND HYPERPARAMETERS FOR $DNN_{NS}$ AND $DNN_S$. $DNN_{NS}$ AND $DNN_S$ ARE THE DNN MODELS FOR NON-SATURATED AND SATURATED PIXELS, RESPECTIVELY

| | $DNN_{NS}$ | | $DNN_S$ | |
|---|---|---|---|---|
| | tested values (range) | final value | tested values (range) | final value |
| Input features | $R_{rc}$ at 412, 443, 469, 488, 531, 547, 555, 645, 667, 678, 748, 859, 869, 1240, 2130 nm | $R_{rc}$ at 412, 443, 488, 531, 547, 667, 678, 748, 869 nm | $R_{rc}$ at 412, 443, 469, 488, 531, 547, 555, 645, 859, 1240, 2130 nm | $R_{rc}$ at 412, 443, 469, 488, 531, 547, 555, 645, 859 nm |
| epochs | 200 | | 150 | |
| hidden layers | 3,5,10,15,20, 25,30,35,40 | 30 | 3,5,10,15 | 5 |
| neurons per layer | 32,64,128,256, 512,1024,2048 | 128 | 32,64,128,256, 512,1024,2048 | 1024 |

3 window, and the window was excluded if it included fewer than five valid pixels. MODIS $R_{rc}$ spectra and $Chla_{OLCI}$ data were extracted from the co-located pixels with CV of $R_{rc}(859) < 40\%$ to obtain more potential matchups (~40% more than using CV < 15%). Additional criteria for non-saturated and saturated pixels were as follows.

To extract the matchups for the non-saturated pixels, the pixels with $Chla_{OLCI} > 100$ mg $m^{-3}$ were excluded first in case there were cyanoHABs of high density. Then the pixels where all ocean bands were not saturated were allowed to remain. Additionally, the image was excluded if the selected pixels were less than half for the entire lake [48]. For saturated pixels, the pixels with $CI_{OLCI} > 0.001$ were screened from those saturated at 667, 678 or 748 nm after excluding the cloud pixels. The non-saturated pixels were collected from 327 images that were consistently distributed across the calendar months, while saturated pixels were collected from 372 images that were predominantly gathered during the wet season months, so the matchups represent the comprehensive conditions as much as possible. The numbers of matched data pairs between OLCI and MODIS were 3 606 420 and 220 584 for non-saturated and saturated pixels, respectively.

*b) DNN models:* DNN models were denoted as $DNN_{NS}$ and $DNN_S$ for non-saturated and saturated pixels, respectively. Both networks were structured and trained following the same workflow. Their input features and the corresponding optimal hyperparameters were tabulated in Table I. They were implemented with the TensorFlow Python API (https://www.tensorflow.org/api_docs/python/tf). $Chla_{OLCI}$ was log-transformed and used as the output.

The networks used the fully connected hidden layers because the data structure of the input feature variables is relatively simple [54]. The rectified linear unit (ReLU) activation function was used for the hidden layers, and a linear activation function was used for the output layer. The Adaptive Moment Estimation (Adam) was employed as optimizer and the mean squared difference (MSD) as the loss function. The matchup datasets were shuffled and split into training, validation, and test sets, and the same sets were used for all trials of either $DNN_{NS}$ or $DNN_S$. The testing set was used to show the model performance and determine the final model. MSD was calculated from the training and validation sets for each epoch, and the comparison between training and validation loss was used to check the stability and overfitting in the epochs. Additionally, the model with the minimal validation MSD during the epochs was saved and its performance on the testing set was compared, representing the best model under the same hyperparameters. The final model, selected from these saved models, refers to the one with the minimal validation MSD and best performance on the testing set.

The single $R_{rc}$ bands served as input features because of the large satellite-satellite matchup datasets [54]. Before feeding to the network, the input $R_{rc}$ spectra were preprocessed sequentially with L2 normalization, min-max scaling, and standardization to make $R_{rc}$ spectrum (each row vector) have a unit norm, linearly scale the individual features down into 0–1, and transfer them to standard normally distributed data, respectively. Based on the previous studies [26], [31], [52], [53], [61], [68], [69] and the model performance on the testing set, $R_{rc}$ at all bands from 412 to 2130 nm and the spectral slopes were tested to determine the optimal input features. And $R_{rc}$ at visible and near-infrared bands were selected for $DNN_{NS}$ and $DNN_S$. $DNN_{NS}$ used all ocean color bands (412, 443, 488, 531, 547, 667, 678, 748, and 869 nm) since adding land bands did not improve the model performance on the testing set. Similar to earlier works of estimating Chla in turbid waters [70], the 667, 678, and 748 bands that provide the essential, consecutive spectral information for red and red-edge domains are used in $DNN_{NS}$ model to improve the model performance. $DNN_S$ used land bands together with the blue and green ocean color bands (412, 443, 469, 488, 531, 547, 555, 645, and 859 nm). In this model, the land bands likely supplement spectral information in the red and near-infrared bands for the saturated pixels.

Table I shows the optimizing range of the hyperparameters. The hyperparameters used in $DNN_{NS}$ and $DNN_S$ included the number of hidden layers and the number of neurons in each layer, which specify the network size and depend on the size of training dataset. The training/validation/test splitting was (50%, 25%, and 25%) and (80%, 10%, and 10%) for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

$DNN_{NS}$ and $DNN_S$, respectively, to make sufficient samples for training and testing. Because of the large sample sizes, when a model with fewer layers/neurons does not possess enough parameters to account for the complexities in the training data, a larger and/or deeper network can be trained without overfitting the data. Thirty hidden layers (128 neurons per layer) and five hidden layers (1024 neurons per layer) were optimal for $DNN_{NS}$ and $DNN_S$, respectively, while the other tested hyperparameters did not show better performance on the testing sets and could be numerically unstable for more epochs.

### *F. Spatial Patterns and Long-Term Changes of $Chla_{MODIS}$*

Monthly $Chla_{MODIS}$ maps were composed from daily $Chla_{MODIS}$ for all months between 2000 and 2023, with mean monthly $Chla_{MODIS}$ was used to extract the long-term trends. When constructing the long-term data series, MODIS/Terra were selected because its duration is longer than MODIS/Aqua. Also, Terra overpass times are within 90 min of OLCI overpass times, thus reducing the impact of vertical and horizontal movement of cyanobacteria blooms [61], [64], [65].

The time series decomposition method in the statsmodels package of Python was used with the additive model. The long-term trend that describes gradual changes in long term (over a time scale of >1 year) was calculated using a centered moving average of 12 months. Furthermore, the significance ($p$-value or $p$) and magnitude ($\beta$) of the trend were obtained by the modified Mann–Kendall test [71] and Sen's Slope method [72], respectively. The modified Mann–Kendall test was conducted with a confidence level of 95% ($\alpha = 0.05$). The Sen's slope (i.e., $\beta$) was calculated as the median of all the slopes estimated between all the successive data points of a time series. A 95% confidence level was used to calculate the upper and lower confidence limits using the procedure described by Gilbert [73].When $p < \alpha$, $\beta > 0$ (or $\beta < 0$) tells that there is a monotonic increasing (or decreasing) trend.

## III. Results

### *A. Performance of Chla Algorithms*

*1) $Chla_{OLCI}$:* Fig. 4 shows the comparisons between in situ Chla and $Chla_{OLCI}$ for the CI model [see (4)], $Chla_{OLCI}$ for the XGB model [see (5)], and the final "hybrid" $Chla_{OLCI}$ determined by combining these values based on the criteria specified in (3). The statistics of the model performances are listed in Table II.

Fig. 4(a) shows relatively high deviation for $Chla_{OLCI}$(CI) (RMSD ~9.2 mg $m^{-3}$ and MAPD ~59.2%). However, this is no surprise because CI does not have an explicit relationship with Chla for $CI_{OLCI} \leq 0$ (the corresponding $Chla_{OLCI}$(CI) < ~20 mg $m^{-3}$). Once again, this highlights the need for a more sophisticated ML method to resolve Chla for low Chla waters. The overall positive correlation between in situ Chla and $Chla_{OLCI}$(CI) ($R^2$ of 0.62), though, indicates that the CI model [see (4)] can be used to determine Chla for $CI_{OLCI} > 0$ waters. Deviations about the best-fit line for $CI_{OLCI} > 0$ can be attributed to the bias in field sampling due to the nonuniformly distributed horizontal and vertical structures of cyanoHABs [64] and differences in in situ-satellite sampling times [65].

TABLE II

Statistics of the Performances for $Chla_{OLCI}$ Determined Using the CI Model (4), $Chla_{OLCI}$ Determined Using the XGB Model (5), and the Final "Hybrid" $Chla_{OLCI}$ Determined by Combining These Values According to the Criteria Specified in (3)

| | CI model (Eq. 4) | XGB model (Eq. 5) | | hybrid model (Eq. 3) |
|---|---|---|---|---|
| | overall | training | testing | overall |
| N | 325 | 189 | 80 | 325 |
| $R^2$ | 0.62 | 0.92 | 0.77 | 0.87 |
| RMSD (mg $m^{-3}$) | 9.2 | 3.1 | 4.1 | 6.9 |
| MAPD (%) | 59.6 | 15.1 | 28.4 | 22.5 |

Fig. 4(b) shows the comparison of in situ Chla and $Chla_{OLCI}$(XGB) for $CI_{OLCI} \leq 0$ waters. The scatter is close to the 1:1 line from ~3 to ~80 mg $m^{-3}$, indicating the trained XGB model works for the optical complexity in LO. $Chla_{OLCI}$(XGB) agrees well with in situ Chla for both training and testing datasets. $R^2$ of training and testing sets are 0.92 and 0.77, respectively. While the RMSD (4.1 mg $m^{-3}$) and MAPD (28.4%) for the testing dataset are a little higher compared to those for the training dataset, the trained XGB model is still robust without overfitting and can be used to derive Chla for $CI_{OLCI} \leq 0$ waters. Fig. 4(c) shows the consistency between in situ Chla and the hybrid $Chla_{OLCI}$ model [see (3)], i.e., the final "hybrid" $Chla_{OLCI}$ product, which combined the scatters of $Chla_{OLCI}$(CI) for $CI_{OLCI} > 0$ and $Chla_{OLCI}$(XGB) for $CI_{OLCI} \leq 0$. The overall performance of the hybrid model shows $R^2$ of ~0.87, RMSD of ~6.9 mg $m^{-3}$, and MAPD of ~22.5%, indicating the effectiveness of the final $Chla_{OLCI}$ product.

Fig. 5 shows representative maps of $Chla_{OLCI}$ and the corresponding true color images. Fig. 5(a) and (b) shows a bloom case in the summer where $CI_{OLCI}$ is mostly greater than 0. The extent of cyanoHABs can be visualized by the greenish features overlying water in the true color image [see Fig. 5(a)] and where $Chla_{OLCI} > 40$ mg $m^{-3}$. Fig. 5(c) and (d) shows a case with turbid, non-bloom waters in the winter where $Chla_{OLCI}$ ranges from <10 to ~20 mg $m^{-3}$. There are no obvious artifacts in the $Chla_{OLCI}$ maps for both cases, indicating the seamless transition from pixels with $CI_{OLCI} > 0$ and $CI_{OLCI} \leq 0$.

*2) ChlaMODIS:* $Chla_{MODIS}$ is compared to $Chla_{OLCI}$ to show the performances of $DNN_{NS}$ and $DNN_S$ on the testing sets. The scatter plots are shown in Fig. 6(a) and (b), with color representing data density, and the statistics are listed in Table III. Although there are some outliers in these scatter-plot comparisons, most values are near the 1:1 line, as shown by the high-density colors. Fig. 6(a) shows that the overall agreement between $Chla_{MODIS}(DNN_{NS})$ and $Chla_{OLCI}$ is close to the 1:1

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

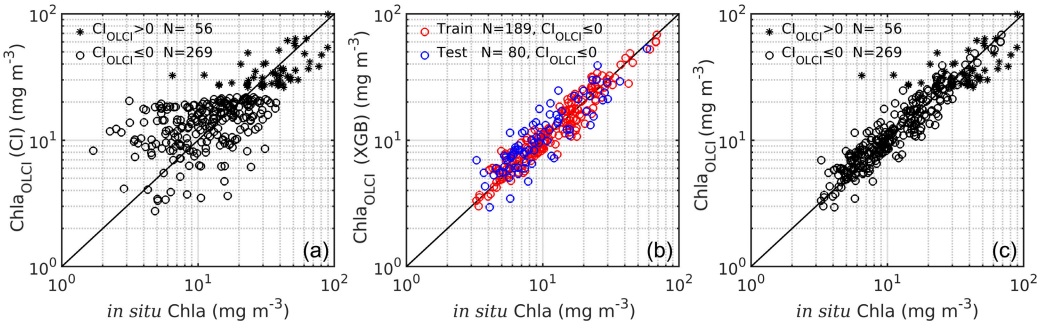



Fig. 4. Comparisons of in situ Chla and $Chla_{OLCI}$ derived using (a) CI model: linear regression between $CI_{OLCI}$ and in situ Chla for $CI_{OLCI} > 0$ [see (4)], (b) XGB model: ML model trained for $CI_{OLCI} \leq 0$ [see (5)], and (c) hybrid model: combined CI and XGB models [see (3)], where CI model was applied to $CI_{OLCI} > 0$ and XGB model was applied to $CI_{OLCI} \leq 0$. The solid black lines represent the 1:1 relationship. The star and circle represent $CI_{OLCI} > 0$ and $CI_{OLCI} \leq 0$ waters, respectively.

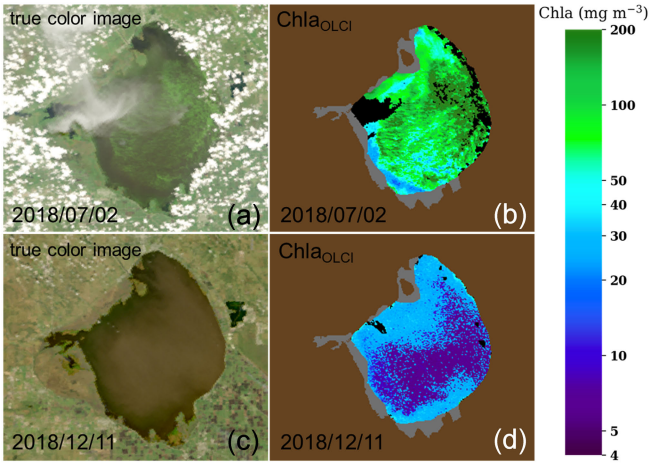



Fig. 5. Representative OLCI images on (a) and (b) July 2, 2018 and (c) and (d) December 11, 2018. (Top and bottom) Panels show the cases with bloom water and turbid, non-bloom waters, respectively. (a) and (c) True-color images, while (b) and (d) corresponding maps of $Chla_{OLCI}$ obtained using the hybrid model [see (3)]. Land and the Littoral zone are in brown. Flagged pixels are black. The gray area is the excluded Nearshore ecological zone (shown in Fig. 1).

TABLE III

STATISTICS OF $CHLA_{MODIS}$ ($DNN_{NS}$) AND $CHLA_{MODIS}$ ($DNN_S$) COMPARED TO $CHLA_{OLCI}$, RESPECTIVELY, AS WELL AS THE FINAL $CHLA_{MODIS}$ COMPARED TO IN SITU CHLA

| | dataset | N | $R^2$ | RMSD (mg m$^{-3}$) | MAPD (%) |
|---|---|---|---|---|---|
| $Chla_{MODIS}$ ($DNN_{NS}$) v.s. $Chla_{OLCI}$ | overall* | 3606420 | 0.93 | 3.9 | 14.0 |
| | training | 2163852 | 0.93 | 3.8 | 13.8 |
| | validation | 721284 | 0.93 | 4.0 | 14.4 |
| | testing | 721284 | 0.93 | 4.0 | 14.4 |
| $Chla_{MODIS}$ ($DNN_S$) v.s. $Chla_{OLCI}$ | overall** | 220584 | 0.89 | 11.3 | 9.7 |
| | training | 176468 | 0.90 | 10.7 | 9.3 |
| | validation | 22059 | 0.85 | 13.6 | 11.2 |
| | testing | 22059 | 0.84 | 13.3 | 11.3 |
| $Chla_{MODIS}$ v.s. *in situ* Chla | | 377 | 0.77 | 6.0 | 24.5 |

Note: * matched at non-saturated pixels; ** matched at saturated pixels.

line from ~3 to ~80 mg m$^{-3}$, with high-density values ranging from ~3 to ~10 mg m$^{-3}$. $DNN_{NS}$ shows $R^2$ of ~0.93, RMSD of < 4.0 mg m$^{-3}$, and MAPD < ~14.4% among the training, validation, and testing sets. $DNN_S$ also shows the consistency between $Chla_{MODIS}$($DNN_S$) and $Chla_{OLCI}$ from ~30 to ~300 mg m$^{-3}$[see Fig. 6(b)], with $R^2$ of 0.84, RMSD of 13.3 mg m$^{-3}$, and MAPD of 11.3% on the testing set. For $DNN_{NS}$ and $DNN_S$, the performance on the testing set is almost the same as the validation set and close to the training set, which indicates the effectiveness and robustness of the trained networks as well as general applicability to the entire lake without overfitting. To validate the final blended $Chla_{MODIS}$ product (i.e., $Chla_{MODIS}$($DNN_{NS}$) for non-saturated pixels and $Chla_{MODIS}$($DNN_S$) for saturated pixels), Fig. 6(c) shows the comparison between in situ Chla and $Chla_{MODIS}$. The data are still close to 1:1 line and show strong correlation ($R^2$ ~ 0.77) for Chla from ~3–~100 mg m$^{-3}$. The point-wise uncertainty for $Chla_{MODIS}$ was characterized as RMSD of ~6.0 mg m$^{-3}$ and MAPD of ~24.5%. Notably, the data pairs matched with in situ Chla are all unseen data for the DNN models. Thus, $Chla_{MODIS}$ is not only consistent with $Chla_{OLCI}$ but also with in situ Chla.

Fig. 7 shows representative images of MODIS/Terra on the same days as shown for the OLCI/S3A images (see Fig. 5). Fig. 7(a) shows the cyanoHABs captured by MODIS in the summer, including saturated pixels in the northeastern portions of the lake caused by high density cyanoHABs where $Chla_{OLCI}$ is close to 200 mg m$^{-3}$ [see Fig. 5(b)]. The true color image in Fig. 7(c) shows turbid waters in the winter where all pixels are non-saturated. $Chla_{MODIS}$ maps show natural variations in spatial distribution, indicating a seamless transition between non-saturated and saturated pixels, and the range and pattern of $Chla_{MODIS}$ are close to those of $Chla_{OLCI}$.

*3) Monthly $Chla_{MODIS}$:* In addition to verifying the daily $Chla_{MODIS}$ using daily $Chla_{OLCI}$ and in situ Chla, the monthly mean $Chla_{MODIS}$ obtained from Terra and Aqua for the entire lake were further compared to monthly mean $Chla_{OLCI}$ obtained from OLCI/S3A in 2016–2023 (see Fig. 8). This is an important comparison to make because the monthly $Chla_{MODIS}$ were used to analyze long-term changes. Overall, both monthly $Chla_{MODIS}$ of Terra and Aqua show consistent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

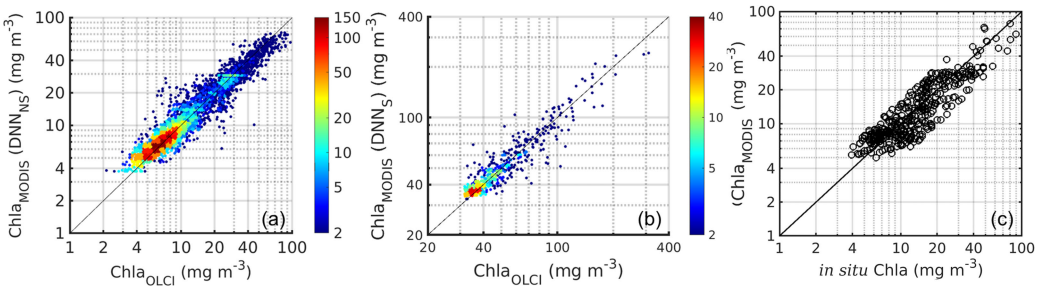



Fig. 6. Comparisons of $Chla_{OLCI}$ with $Chla_{MODIS}$ on the testing set for (a) $DNN_{NS}$ and (b) $DNN_S$. (c) Comparison of in situ Chla and $Chla_{MODIS}$. The color in the scatter plot represents the density of the points. The solid black lines represent the 1:1 relationship.

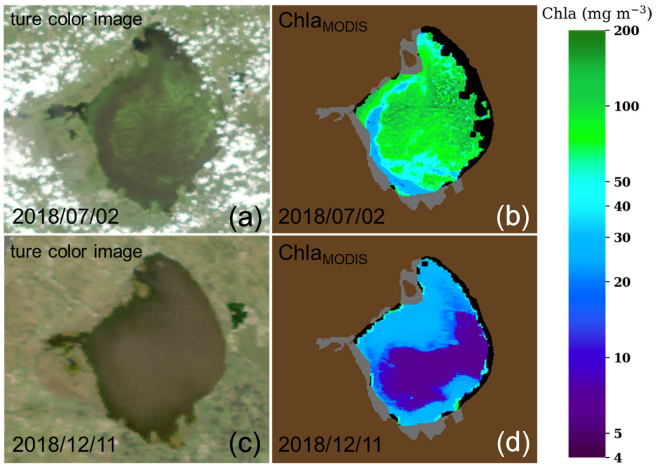



Fig. 7. Representative images of MODIS/Terra on the same days shown in Fig. 5. Top and bottom panels show the cases with bloom water and turbid, non-bloom waters, respectively. (a) and (c) True color images. (b) and (d) Maps of the final $Chla_{MODIS}$ obtained by combining $Chla_{MODIS}$ ($DNN_{NS}$) and $Chla_{MODIS}$ ($DNN_S$). Land and littoral zone are in brown. Flagged pixels are black. The gray area is the excluded nearshore ecological zone (shown in Fig. 1).

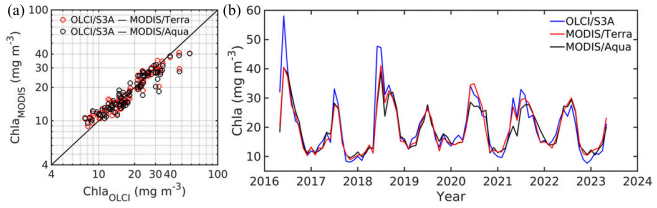



Fig. 8. (a) Comparison of monthly mean $Chla_{OLCI}$ and $Chla_{MODIS}$ for the entire lake in 2016–2023. (b) Time series of monthly mean $Chla_{MODIS}$ and $Chla_{OLCI}$ for the entire lake in 2016–2023. $Chla_{MODIS}$ was obtained from both Terra and Aqua and $Chla_{OLCI}$ was obtained from S3A.

temporal variations as with monthly $Chla_{OLCI}$ for the range of 10–40 mg $m^{-3}$. For the time series [see Fig. 8(b)], typical summer peaks in Chla when cyanoHABs were often prevalent and winter troughs in Chla when conditions are less favorable for bloom formation were observed for both OLCI and MODIS. The slightly different magnitudes can be attributed to the different overpass dates in a month (i.e., temporal aliasing) and changes in waters and/or clouds on the same day even when data were collected within ~90 min (e.g., MODIS/Terra). Fig. 8(a) shows that monthly $Chla_{MODIS}$ obtained from both Terra and Aqua agree well with monthly $Chla_{OLCI}$. For Terra and Aqua, respectively, $R^2$ were 0.90 and 0.87, RMSD were 3.7 and 4.3 mg $m^{-3}$, and MAPD were 11.3% and 13.2%. Based on these results, we inferred that the proposed Chla inversion method for MODIS is reliable and can therefore be used to examine spatiotemporal variability and assess long-term change in LO.

### B. Spatial and Temporal Patterns of $Chla_{MODIS}$

*1) Monthly and Climatology Mean:* Monthly $Chla_{MODIS}$ obtained from MODIS/Terra in 2000–2023 was used to assess the spatiotemporal changes of $Chla_{MODIS}$ in LO. Fig. 9(a) shows that mean $Chla_{MODIS}$ averaged across the entire lake varied from ~10 to ~40 mg $m^{-3}$ with the local minima and maxima generally coinciding with winter (i.e., dry – non-bloom) and summer (i.e., wet – bloom) seasons, respectively. $Chla_{MODIS}$ peaks up to ~40 mg $m^{-3}$ were observed in several years with intensive cyanoHABs events, such as 2005, 2016, 2018, and 2023. Fig. 9(b) shows the monthly climatology maps of $Chla_{MODIS}$ in LO. Generally, $Chla_{MODIS}$ in the entire lake starts to gradually increase in spring (March–May), attains maximal values in the summer (June–August), declines gradually in autumn (September–November), and exhibits minimal values in the winter (December–February). Clear spatial patterns of $Chla_{MODIS}$ in LO were also evident from the climatology maps. $Chla_{MODIS}$ in the Northern and Western zones of LO were generally higher than $Chla_{MODIS}$ in the Central zone, which could be attributed to reduced light limitation and increased nutrient inflows in these areas of the lake [19], [37], [39].

The spatial distribution patterns, seasonality, and long-term changes of Chla revealed in Figs. 8(b) and 9, together with the validation results, suggest that the proposed method is applicable to both bloom waters and turbid, non-bloom waters throughout the year.

*2) Long-Term Trends:* The long-term trend components of $Chla_{MODIS}$ ($Chla_t$) obtained from MODIS/Terra over the past 20+ years (2000–2023) were decomposed from the monthly time series of $Chla_{MODIS}$ for the entire lake and the three major ecological zones (Northern, Central, Western) [37] (see Fig. 10). Table IV shows the $p$-value of $Chla_t$ is always less than 0.001, indicating that the observed positive trends (~2 mg $m^{-3}$ $decade^{-1}$) for the entire lake and each individual zone are statistically significant. The percentage increase of $Chla_{MODIS}$ was ~30% over the past 20 years, as compared to the multiple year average. $Chla_t$ increased at a faster

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

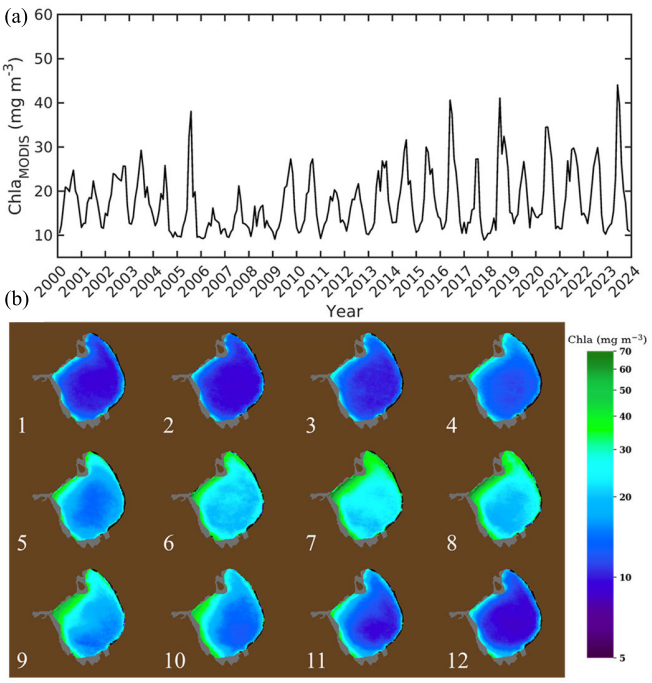



Fig. 9. (a) Time series of monthly mean $Chla_{MODIS}$ in 2000–2023 obtained from MODIS/Terra. (b) Monthly Climatology $Chla_{MODIS}$ maps. Land and littoral zone are in brown. The gray area is the excluded nearshore ecological zone (shown in Fig. 1).

TABLE IV

STATISTICS OF LONG-TERM TRENDS AND THE PERCENTAGE INCREASES OF $Chla_{MODIS}$ OVER THE PAST 20+ YEARS (2000–2023) IN THE ENTIRE LAKE AND FOR MAJOR ECOLOGICAL ZONES

| | LO | Northern | Central | Western |
|---|---|---|---|---|
| p-value | <0.001 | <0.001 | <0.001 | <0.001 |
| slope (mg $m^{-3}$ $decade^{-1}$) | 2.1 | 2.9 | 1.9 | 2.3 |
| percentage increase (%) | 27.5 | 33.9 | 31.3 | 27.4 |

rate for the Northern and Western ecological zones (2.9 and 2.3 mg $m^{-3}decade^{-1}$, respectively) where river/canal input (i.e., nutrient loading) is highest, compared to the Central ecological zone (1.9 mg $m^{-3}decade^{-1}$) where light limitation associated with the increased likelihood of wind-driven suspended sediments typically limits phytoplankton growth [19], [37].

Major peaks in $Chla_{MODIS}$ ($Chla_t$) over the past 20+ years are often associated with major El Niño events (e.g., 2002–2003, 2009–2010, and 2015–2016) when wetter than normal dry seasons in South Florida tend to occur [40] and extreme weather events such as active or strong hurricane seasons (e.g., 2004–2005, 2017, and 2022) [74]. The decline in $Chla_{MODIS}$ that occurred for several years following the strong 2004–2005 hurricane season may be attributed to a regime shift where diatoms instead of cyanobacteria dominated phyto-

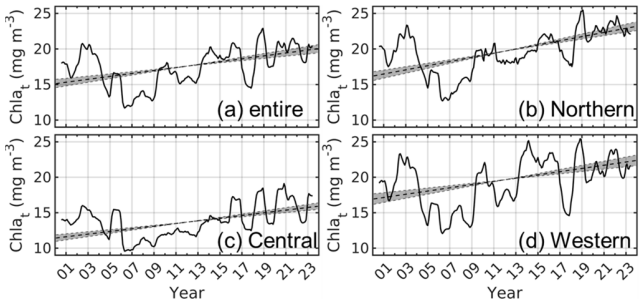



Fig. 10. Long-term trend component of $Chla_{MODIS}$ ($Chla_t$) in (a) entire lake and (b)–(d) major ecological zones (Northern, Central, and Western). Dashed lines are linear trend-fitting lines, and the gray areas show the confidence intervals of these trends with $\alpha = 0.05$.

plankton populations and blooms generally did not occur likely because of elevated total suspended solids and lower water transparency [75], [76].

## IV. DISCUSSION

### A. ML: Direct or Indirect?

Due to a finite number of stations sampled once or twice monthly, in situ water quality sampling conducted as part of routine monitoring programs for LO cannot adequately track responses of Chla to targeted nutrient reductions [5] and fails to capture the spatiotemporal variability of bloom events [23]. Remotely sensed Chla is well-poised to meet these observational needs and can provide managers with more timely and robust information for improved decision making. However, while remotely sensed Chla determined using the widely used CI model [33], [41] provides a robust indicator of bloom occurrence, background Chla (i.e., non-bloom conditions) cannot be reliably determined, thus hindering the assessment of long-term trends. Previous attempts to use satellite data to estimate Chla in LO were either through demonstration [31] or the use of limited MERIS or OLCI data that discarded low-CI values (thus focused on "bloom" only) [23], [34], [35], [36]. In this study, a new ML model was developed and applied to yield a continuous long-term MODIS Chla data record (2000–2023) that is robust over a wide dynamic range (~4–100 mg $m^{-3}$), and thus optimal for assessing long-term trends in support of ongoing management efforts. Because of lack of long-term Chla data in most turbid lakes worldwide, the approach leading to such a model, although developed for LO, is expected to be widely applicable.

The novelty of the model design in this work lies in the use of $Chla_{OLCI}$ as a "bridge" to avoid the problem of data scarcity encountered when in situ data were used to "directly" train the ML model. Although several MODIS Chla algorithms have been developed using in situ data for other lakes [26], [29], [68], this study found that training a model directly through in situ data does not work well for LO. Here, a case study of utilizing strength of ML and multiple sensors to estimate Chla for MODIS imagery is presented for LO. Compared to MODIS, OLCI carries a 709 nm band that is useful for estimating Chla in turbid optically complex waters [67]. Indeed, this band contributed to two of the most important band combinations used as input features, allowing for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

XGB model (based on matching pairs of OLCI and in situ Chla data) to train well. Also, OLCI does not exhibit signal saturation over high density cyanoHABs where MODIS ocean bands often saturate [41]. Thus, $Chla_{OLCI}$ consistent to in situ Chla can serve as the ground truth to generate the infinitely large OLCI-MODIS matchup datasets for developing the DNN models to indirectly retrieve $Chla_{MODIS}$. The large size and diversity of this new training dataset is nearly impossible to recreate using in situ Chla match-ups alone (i.e., direct approach), thus ensuring adequate training of the NNs [54]. Additionally, the proposed method of utilizing intermediate $Chla_{OLCI}$ to indirectly estimate $Chl_{MODIS}$ can provide consistent Chla estimations between MODIS and OLCI, which will ensure a continuous long-term Chla data record for monitoring change beyond the lifespan of MODIS.

Other studies have introduced similar multi-sensor approaches to provide seamless long-term satellite data records across multiple missions [27], [48], [61]. However, few employed ML approaches to align multi-mission Chla estimations over optically complex waters and instead utilized standard empirical or semi-analytical models that often require limited bands. Studies that invoked the powers of both multi-sensors and ML were often focused on generating a long-term timeseries for monitoring blooms only. Here, we demonstrate the power of combining both multi-sensor and more powerful ML approaches to yield accurate Chla across multiple years and for various optical water types, including bloom and turbid, non-bloom waters.

### B. Model Uncertainties

The point-wise uncertainty was evaluated using in situ Chla for daily $Chla_{OLCI}$ (RMSD of ~6.9 mg $m^{-3}$ and MAPD of ~22.5%) and $Chla_{MODIS}$ (RMSD of ~6.0 mg $m^{-3}$ and MAPD of ~24.5%), respectively. The uncertainty of Chla estimates was proposed to be within 70% for optically complex waters [77], where the published uncertainty for lake waters varied from ~30% to >70% [25], [51], [70] and often exceeded 40% [78], [79]. As such, although the uncertainties of Chla estimates in this study appear high, they are actually lower than most of the published values. The bias inherent with field sampling of cyanoHABs [64] is one of the uncertainty sources, and so is the similarity in $R_{rc}$ spectra for the non-bloom optically complex waters [51]. MAPD of daily $Chla_{MODIS}$ is slightly larger than that of $Chla_{OLCI}$ due to the effect of error propagation, which may be also relevant to the changes in "ground truth" because the nonuniformly distributed cyanoHABs can change between the overpass times of MODIS and OLCI. Also, the difference in spatial resolution of original imagery (300-m OLCI and 1-km MODIS) could introduce uncertainty of observation, even though they have been reprojected into the same grid at 250 m. Notably, for the monthly $Chla_{MODIS}$ used for analysis, the uncertainty characterized using monthly $Chla_{OLCI}$ shows an RMSD of 3.7 mg $m^{-3}$ and MAPD of 11.3%. The relatively low MAPD in both daily and monthly $Chla_{MODIS}$ (~25% and ~11%) suggests the derived product is robust and reliable enough to analyze the long-term trend of Chla. Additionally, the magnitude of monthly mean $Chla_{MODIS}$ is often relatively low (~10–40 mg $m^{-3}$) compared to the extremely high biomass (Chla > 100 mg $m^{-3}$) blooms observed in the daily maps. This is because the intense blooms are often only visible a few times per month when wind stress is low [80], and they are effectively averaged out with the non-bloom days, which is more representative of the actual conditions.

However, these observations do not indicate there is no room for further improvement. Given the known uncertainty sources, the model performance could be improved in the future through increasing the sampling frequency and spatial coverage of field measurements and through improving atmospheric correction so the model can be developed from satellite-derived $R_{rs}$ instead of $R_{rc}$. Similarly, because the new Ocean Color Instrument (OCI) of the newest NASA PACE mission is a hyperspectral sensor that is expected to provide superior performance to other sensors, the use of OCI instead of OLCI as a bridge to MODIS may also improve the model performance.

Nevertheless, even with the known uncertainties, the models proposed in this study show much improved performance over the published models because those models were designed for waters with different optical complexity [51], [52]. Furthermore, the proposed models have been shown to be applicable to the entire lake in all seasons (see Tables II and III), and they may be applicable to other lakes as long as the optical complexity is similar. Even if the exact models are not directly applicable to other lakes, the general approach of this study (i.e., the use of OLCI as a bridge between in situ and MODIS measurements) should be generally applicable to other coastal or inland waters once local in situ data are available to retrain the models.

### C. Spatial and Temporal Patterns

The spatial pattern of the $Chla_{MODIS}$ climatology in summer is shown in Fig. 11, where the black dash lines indicate the $Chla_{MODIS}$ contours of 21, 25, and 30 mg $m^{-3}$. This spatial pattern observed from $Chla_{MODIS}$ was further compared to the boundaries of the major ecological zones defined by Phlips et al. [37] (red dashed lines in Fig. 11). These latter ecological zones were based on a more comprehensive in situ water quality dataset (e.g., nutrients, Chla, total suspended sediments, and light availability) collected over a two-year period (1988–1989). Nevertheless, both the in situ and remote sensing results show the tendency for higher Chla waters ($Chla_{MODIS}$ > 40 mg $m^{-3}$) to occur in the Western and Northern zones where nutrient influx is highest and light limitation is reduced. This reinforces the potential risks of discharging toxic bloom-laden water with high Chla west of LO to the Caloosahatchee River Estuary in the summer when blooms are present [81]. The area with relatively low $Chla_{MODIS}$ (<21 mg $m^{-3}$) matches somewhat closely with the boundary of the Central ecological zone and indicates the area where phytoplankton growth is generally controlled by light limitation due to high turbidity [32], [37]. More in-depth ecological interpretation in terms of Chla spatial patterns and zoning requires further analysis in the future when other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

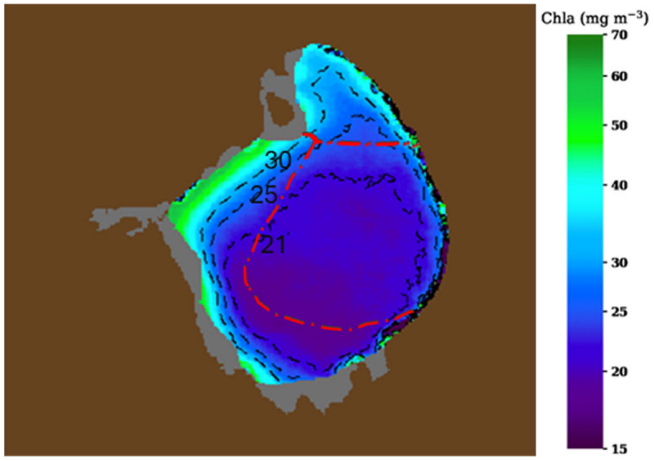



Fig. 11. $Chla_{MODIS}$ climatology map in the summer (June–August). Land and littoral zone are in brown. The gray area is the excluded Nearshore ecological zone. Black dashed lines represent $Chla_{MODIS}$ contours of 21, 25, and 30 mg m$^{-3}$. The red dashed lines show the boundaries of the Central, Northern, and Western ecological zones as defined in Phlips et al. [37] (shown in Fig. 1).

data (e.g., nutrient concentrations and fluxes) on the same observational scale become available.

In terms of temporal patterns, $Chla_{MODIS}$ is generally higher in the summer when conditions are more favorable for blooms (e.g., higher temperatures, longer days with increased solar radiation, and lower winds) [82] compared to winter when the frequent passage of frontal systems leads to increased turbidity thus suppressing blooms [32]. Overall, $Chla_{MODIS}$ has increased by ~30% over the past 20+ years in LO with more rapid increases occurring in the Northern and Western ecological zones (2.9 and 2.3 mg m$^{-3}$ decade$^{-1}$, respectively) compared to the Central zone (1.9 mg m$^{-3}$ decade$^{-1}$). Compared to the Central zone where light availability limits algal growth all year around, the Northern and Western zones experience nutrient (N or P) limitation in the summer [19]. This suggests that future efforts geared toward combining remotely sensed Chla with in situ nutrient data collected as part of routine monitoring programs may help identify hot spots where Chla displays a higher sensitivity to nutrient loading, thus informing managers with regard to eutrophication management strategies [19].

Aside from nutrients, several other environmental factors may be responsible for driving the long-term trend of $Chla_{MODIS}$. These were evaluated using multi-variate analysis. The correlation coefficients ($r$) of each factor with $Chla_{MODIS}$ (see Table V) were determined on a monthly, annual, summertime, and wintertime basis. Annual means were determined based on water-years that begin in May of the previous year and end in April, thereby encompassing an entire wet and dry season. SST shows strong positive correlations with $Chla_{MODIS}$ over monthly and annual time scales ($r = 0.70$ and 0.58, respectively). This is likely because high temperatures favor cyanobacteria blooms through increased growth rates and water column stability [82]. Using MODIS SST (2000–2021), Shi and Hu [7] showed that LO is warming ten times faster than the global ocean (0.80 °C versus 0.08 °C decade$^{-1}$). Lake stage and precipitation also show positive correlations with $Chla_{MODIS}$, with lake stage having a correlation ($r$ = 0.65) in the summertime and precipitation having a correlation ($r = 0.56$) in the wintertime. The difference in correlation of stage and precipitation indicates the different processes affecting nutrient loads during the summer and winter months. The summertime blooms get worse when lake stage is high due to lots of nutrient-rich inflow during the wet season. For the wintertime, MEI shows the correlation of 0.40 with $Chla_{MODIS}$ and strong correlation of 0.72 with precipitation (not shown), indicating the global climate changes impact the wintertime blooms by rainfall during the dry season. Noting that multiple regression model residuals are higher in wintertime than summertime, additional factors (beyond those tested) may be affecting temporal changes of Chla in LO, which requires further analysis in the future.

TABLE V
CORRELATION COEFFICIENTS ($R$) OF VARIOUS ENVIRONMENTAL FACTORS WITH $Chla_{MODIS}$ DETERMINED MONTHLY, ANNUALLY MEAN, SUMMERTIME MEAN (JUNE–AUGUST), AND WINTERTIME MEAN (DECEMBER–FEBRUARY)

| | Monthly | Annual* | Summer | Winter |
|---|---|---|---|---|
| SST | **0.70** | **0.58** | 0.38 | 0.29 |
| Precipitation | 0.46 | 0.33 | -0.17 | **0.56** |
| Inflow | 0.40 | 0.36 | 0.47 | 0.27 |
| Stage | 0.05 | 0.46 | **0.65** | -0.08 |
| MEI | 0.04 | 0.21 | 0.14 | 0.40 |

* Note, "annual" here refers to a water-year (i.e., from May of previous year to April of current year).

## V. CONCLUSION

This study has shown a practical approach to combine in situ observations, multi-sensor remote sensing data, and ML to address the challenges of Chla remote sensing in optically complex inland waters of LO. The approach led to the development of a long-term (24 years) MODIS-based data record to examine spatiotemporal patterns and long-term changes of $Chla_{MODIS}$ in this largest lake of Florida (USA). Increasing Chla trends were revealed in all lake segments, with an average trend of ~2 mg m$^{-3}$ decade$^{-1}$ between 2000 and 2023. The spatial patterns of $Chla_{MODIS}$ generally agreed well with ecological zonation determined previously from in situ water assessments [37]. These data form the basis for future efforts to interpret the driving factors leading to the long-term changes, and will assist in water management efforts.

We attribute these findings to an "indirect" use of in situ data to develop the $Chla_{MODIS}$ model, which leveraged a well-trained $Chla_{OLCI}$ model as the "bridge" to avoid the problem of in situ data scarcity. Such data scarcity made it impossible to develop a data-rich DNN model, but the use of the intermediate $Chla_{OLCI}$ avoided such a problem, thus leading to a more robust remote sensing model. The approach therefore takes advantage of more spectral bands and increased radiometric range of OLCI and longer-term time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

series of MODIS when in situ data are not sufficient to train the data-rich DNN model. The final model is applicable to lake waters under both high algal biomass and low-Chla (i.e., high turbidity) conditions. Because many lakes face the same situation of in situ data scarcity and because of the global availability of both OLCI and MODIS data, we believe that such an approach to use OLCI as a "bridge" to bring limited in situ data and the wealthy MODIS data together should be extendable to other lakes to study long-term Chla changes.

## ACKNOWLEDGMENT

The authors thank Zachariah Welch of the South Florida Water Management District (SFWMD) for the help with interpreting the environmental driving factors, the SFWMD for providing in situ data, and NASA Ocean Biology Distributed Active Archive Center (OB.DAAC) for providing satellite data. They are grateful to the NASA Ocean Biology Processing Group for providing the data processing software. They also thank the three anonymous reviewers for their comments to improve the presentation of this work. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the author(s) and do not necessarily reflect those of NOAA or SFWMD.

## REFERENCES

[1] J. C. Ho, A. M. Michalak, and N. Pahlevan, "Widespread global increase in intense lake phytoplankton Blooms since the 1980s," *Nature*, vol. 574, no. 7780, pp. 667–670, Oct. 2019.

[2] G. Yu et al., "Effects of nitrogen and phosphorus on chlorophyll a in lakes of China: A meta-analysis," *Environ. Res. Lett.*, vol. 17, no. 7, Jul. 2022, Art. no. 074038.

[3] D. R. Engstrom, S. P. Schottler, P. R. Leavitt, and K. E. Havens, "A reevaluation of the cultural eutrophication of lake Okeechobee using multiproxy sediment records," *Ecological Appl.*, vol. 16, no. 3, pp. 1194–1206, Jun. 2006.

[4] K. E. Havens and D. E. Gawlik, "Lake Okeechobee conceptual ecological model," *Wetlands*, vol. 25, no. 4, pp. 908–925, Dec. 2005.

[5] Z. A. Siders and K. E. Havens, "Revisiting the total maximum daily load total phosphorus goal in lake Okeechobee," *Hydrobiologia*, vol. 847, no. 20, pp. 4221–4232, Nov. 2020.

[6] H. W. Paerl and M. A. Barnard, "Mitigating the global expansion of harmful cyanobacterial Blooms: Moving targets in a human- and climatically-altered world," *Harmful Algae*, vol. 96, Jun. 2020, Art. no. 101845.

[7] J. Shi and C. Hu, "South Florida estuaries are warming faster than global oceans," *Environ. Res. Lett.*, vol. 18, no. 1, Jan. 2023, Art. no. 014003.

[8] K. E. Havens, R. T. James, T. L. East, and V. H. Smith, "N: P ratios, light limitation, and cyanobacterial dominance in a subtropical lake impacted by non-point source nutrient pollution," *Environ. Pollut.*, vol. 122, no. 3, pp. 379–390, Apr. 2003.

[9] O. M. Tarabih, T. D. Dang, R. Paudel, and M. E. Arias, "Lake operation optimization of nutrient exports: Application of phosphorus control in the largest subtropical lake in the United States," *Environ. Model. Softw.*, vol. 160, Feb. 2023, Art. no. 105603.

[10] B. J. Kramer et al., "Nitrogen limitation, toxin synthesis potential, and toxicity of cyanobacterial populations in lake Okeechobee and the St. Lucie river estuary, Florida, during the 2016 state of emergency event," *PLoS ONE*, vol. 13, no. 5, May 2018, Art. no. e0196278.

[11] J. S. Metcalf et al., "Public health responses to toxic cyanobacterial Blooms: Perspectives from the 2016 Florida event," *Water Policy*, vol. 20, no. 5, pp. 919–932, Oct. 2018.

[12] *Total Maximum Daily Load (TMDL) Development for Lake Okeechobee, Florida*, USEPA, Atlanta, GA, USA, 2000.

[13] *Total Maximum Daily Load for Total Phosphorus Lake Okeechobee, Florida*, FDEP, Tallahassee, FL, USA, 2001.

[14] D. E. Canfield, R. W. Bachmann, and M. V. Hoyer, "Restoration of lake Okeechobee, Florida: Mission impossible?" *Lake Reservoir Manage.*, vol. 37, no. 1, pp. 95–111, Jan. 2021.

[15] J. N. Boyer, C. R. Kelble, P. B. Ortner, and D. T. Rudnick, "Phytoplankton Bloom status: Chlorophyll a biomass as an indicator of water quality condition in the southern estuaries of Florida, USA," *Ecological Indicators*, vol. 9, no. 6, pp. S56–S67, Nov. 2009.

[16] J. G. Ferreira et al., "Overview of eutrophication indicators to assess environmental status within the European marine strategy framework directive," *Estuarine, Coastal Shelf Sci.*, vol. 93, no. 2, pp. 117–131, Jun. 2011.

[17] K. C. Fickas, R. E. O'Shea, N. Pahlevan, B. Smith, S. L. Bartlett, and J. L. Wolny, "Leveraging multimission satellite data for spatiotemporally coherent cyanoHAB monitoring," *Frontiers Remote Sens.*, vol. 4, pp. 1–24, Sep. 2023.

[18] A. Filazzola et al., "A database of chlorophyll and water chemistry in freshwater lakes," *Sci. Data*, vol. 7, no. 1, p. 310, Sep. 2020.

[19] T. Xu, T. Yang, X. Zheng, Z. Li, and Y. Qin, "Growth limitation status and its role in interpreting chlorophyll a response in large and shallow lakes: A case study in lake Okeechobee," *J. Environ. Manage.*, vol. 302, Jan. 2022, Art. no. 114071.

[20] B. A. Schaeffer, J. D. Hagy, and R. P. Stumpf, "Approach to developing numeric water quality criteria for coastal waters: Transition from SeaWiFS to MODIS and MERIS satellites," *J. Appl. Remote Sens.*, vol. 7, no. 1, Jun. 2013, Art. no. 073544.

[21] K. E. Havens and W. W. Walker, "Development of a total phosphorus concentration goal in the TMDL process for lake Okeechobee, Florida (USA)," *Lake Reservoir Manage.*, vol. 18, no. 3, pp. 227–238, Sep. 2002.

[22] D. E. Canfield Jr., R. W. Bachmann, and M. V. Hoyer, "Long-term chlorophyll trends in Florida lakes," *J. Aquat. Plant Manage*, vol. 56, pp. 47–56, Jun. 2018.

[23] N. Reynolds, B. A. Schaeffer, L. Guertault, and N. G. Nelson, "Satellite and in situ cyanobacteria monitoring: Understanding the impact of monitoring frequency on management decisions," *J. Hydrol.*, vol. 619, Apr. 2023, Art. no. 129278.

[24] Z. Cao et al., "A decade-long chlorophyll—A data record in lakes across China from VIIRS observations," *Remote Sens. Environ.*, vol. 301, Feb. 2024, Art. no. 113953.

[25] Q. Guan, L. Feng, X. Hou, G. Schurgers, Y. Zheng, and J. Tang, "Eutrophication changes in fifty large lakes on the Yangtze plain of China derived from MERIS and OLCI observations," *Remote Sens. Environ.*, vol. 246, Sep. 2020, Art. no. 111890.

[26] J. Li et al., "Estimation of chlorophyll—A concentrations in a highly turbid eutrophic lake using a classification-based MODIS land-band algorithm," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 3769–3783, Oct. 2019.

[27] X. Liu, M. Warren, N. Selmes, and S. G. H. Simis, "Quantifying decadal stability of lake reflectance and chlorophyll—A from medium-resolution ocean color sensors," *Remote Sens. Environ.*, vol. 306, May 2024, Art. no. 114120.

[28] B. Smith et al., "A chlorophyll—A algorithm for Landsat-8 based on mixture density networks," *Frontiers Remote Sens.*, vol. 1, pp. 1–17, Feb. 2021.

[29] Y. Zhang, R. Ma, H. Duan, S. Loiselle, M. Zhang, and J. Xu, "A novel MODIS algorithm to estimate chlorophyll a concentration in eutrophic turbid lakes," *Ecological Indicators*, vol. 69, pp. 138–151, Oct. 2016.

[30] Z. Cao et al., "Landsat observations of chlorophyll—A variations in lake Taihu from 1984 to 2019," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 106, Feb. 2022, Art. no. 102642.

[31] N.-B. Chang, Y. J. Yang, A. Daranpob, K.-R. Jin, and T. James, "Spatiotemporal pattern validation of chlorophyll—A concentrations in lake Okeechobee, Florida, using a comparative MODIS image mining approach," *Int. J. Remote Sens.*, vol. 33, no. 7, pp. 2233–2260, Apr. 2012.

[32] M. Wang, C. J. Nim, S. Son, and W. Shi, "Characterization of turbidity in Florida's lake Okeechobee and caloosahatchee and St. Lucie estuaries using MODIS-Aqua measurements," *Water Res.*, vol. 46, no. 16, pp. 5410–5422, Oct. 2012.

[33] T. T. Wynne et al., "Relating spectral shape to cyanobacterial Blooms in the Laurentian great lakes," *Int. J. Remote Sens.*, vol. 29, no. 12, pp. 3665–3672, Jun. 2008.

[34] J. M. Clark et al., "Satellite monitoring of cyanobacterial harmful algal Bloom frequency in recreational waters and drinking water sources," *Ecol. Indicators*, vol. 80, pp. 84–95, Sep. 2017.

[35] E. A. Urquhart, B. A. Schaeffer, R. P. Stumpf, K. A. Loftin, and P. J. Werdell, "A method for examining temporal changes in cyanobacterial harmful algal Bloom spatial extent using satellite remote sensing," *Harmful Algae*, vol. 67, pp. 144–152, Jul. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.

[36] S. Mishra, R. P. Stumpf, B. A. Schaeffer, P. J. Werdell, K. A. Loftin, and A. Meredith, "Measurement of cyanobacterial Bloom magnitude using satellite remote sensing," *Sci. Rep.*, vol. 9, no. 1, p. 18310, Dec. 2019.

[37] E. J. Phlips et al., "Spatial and temporal variability of trophic state parameters in a shallow subtropical lake (Lake Okeechobee, Florida, USA)," *Archiv fürobiologie*, vol. 128, no. 4, pp. 437–458, Nov. 1993.

[38] L. Lai et al., "Satellite mapping reveals phytoplankton biomass's spatio-temporal dynamics and responses to environmental factors in a eutrophic inland lake," *J. Environ. Manage.*, vol. 360, Jun. 2024, Art. no. 121134.

[39] S. Faridmarandi, Y. P. Khare, and G. M. Naja, "Long-term regional nutrient contributions and in-lake water quality trends for lake Okeechobee," *Lake Reservoir Manage.*, vol. 37, no. 1, pp. 77–94, Jan. 2021.

[40] N. Schmidt, E. K. Lipp, J. B. Rose, and M. E. Luther, "ENSO influences on seasonal rainfall and river discharge in Florida," *J. Climate*, vol. 14, no. 4, pp. 615–628, Feb. 2001.

[41] T. T. Wynne, R. P. Stumpf, and T. O. Briggs, "Comparing MODIS and MERIS spectral shapes for cyanobacterial Bloom detection," *Int. J. Remote Sens.*, vol. 34, no. 19, pp. 6668–6678, Oct. 2013.

[42] C. Hu, "A novel ocean color index to detect floating algae in the global oceans," *Remote Sens. Environ.*, vol. 113, no. 10, pp. 2118–2129, 2009.

[43] D. Zhao, L. Feng, and K. Sun, "Development of a practical atmospheric correction algorithm for inland and nearshore coastal waters," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5402515.

[44] M. Shen et al., "Random forest: An optimal chlorophyll—A algorithm for optically complex inland water suffering atmospheric correction uncertainties," *J. Hydrol.*, vol. 615, Dec. 2022, Art. no. 128685.

[45] OBPG. (Jul. 24, 2023). *CyAN Reprocessing Version 5*. [Online]. Available: https://oceancolor.gsfc.nasa.gov/data/reprocessing/projects/cyan/version/5/

[46] OBPG. (Jul. 24, 2024). *Level 2 Ocean Color Flags*. [Online]. Available: https://oceancolor.gsfc.nasa.gov/resources/atbd/ocl2flags/

[47] M. Cichra, S. Badylak, N. Henderson, B. Rueter, and E. Phlips, "Phytoplankton community structure in the open water zone of a shallow subtropical lake (Lake Okeechobee, Florida, USA)," *Arch. Hydrobiol. Adv. Limnol.*, vol. 45, no. 45, pp. 157–176, 1995.

[48] T. T. Wynne, S. Mishra, A. Meredith, R. W. Litaker, and R. P. Stumpf, "Intercalibration of MERIS, MODIS, and OLCI satellite imagers for construction of past, present, and future cyanobacterial biomass time series," *Remote Sens.*, vol. 13, no. 12, p. 2305, Jun. 2021.

[49] M. C. Tomlinson et al., "Relating chlorophyll from cyanobacteria-dominated inland waters to a MERIS Bloom index," *Remote Sens. Lett.*, vol. 7, no. 2, pp. 141–149, Feb. 2016.

[50] Z. Cao et al., "A machine learning approach to estimate chlorophyll—A from Landsat-8 measurements in inland lakes," *Remote Sens. Environ.*, vol. 248, Oct. 2020, Art. no. 111974.

[51] N. Pahlevan et al., "Seamless retrievals of chlorophyll—A from Sentinel-2 (MSI) and Sentinel-3 (OLCI) in inland and coastal waters: A machine-learning approach," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111604.

[52] Z. Cao et al., "What water color parameters could be mapped using MODIS land reflectance products: A global evaluation over coastal and inland waters," *Earth-Sci. Rev.*, vol. 232, Sep. 2022, Art. no. 104154.

[53] L. Lai, Y. Zhang, Z. Cao, Z. Liu, and Q. Yang, "Algal biomass mapping of eutrophic lakes using a machine learning approach with MODIS images," *Sci. Total Environ.*, vol. 880, Jul. 2023, Art. no. 163357.

[54] G. Zheng, C. W. Brown, and P. M. DiGiacomo, "Retrieval of oceanic chlorophyll concentration from GOES-R advanced baseline imager using deep learning," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113660.

[55] G. Richardson, N. Foreman, A. Knudby, Y. Wu, and Y. Lin, "Global deep learning model for delineation of optically shallow and optically deep water in Sentinel-2 imagery," *Remote Sens. Environ.*, vol. 311, Sep. 2024, Art. no. 114302.

[56] D. Perez et al., "Quantifying seagrass distribution in coastal water with deep learning models," *Remote Sens.*, vol. 12, no. 10, p. 1581, May 2020.

[57] H. Guo, X. Zhu, J. Jeanne Huang, Z. Zhang, S. Tian, and Y. Chen, "An enhanced deep learning approach to assessing inland lake water quality and its response to climate and anthropogenic factors," *J. Hydrol.*, vol. 620, May 2023, Art. no. 129466.

[58] V. Sagan et al., "Monitoring inland water quality using remote sensing: Potential and limitations of spectral indices, bio-optical simulations, machine learning, and cloud computing," *Earth-Sci. Rev.*, vol. 205, Jun. 2020, Art. no. 103187.

[59] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[60] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, p. 111716, May 2020.

[61] S. Mishra, R. P. Stumpf, and A. Meredith, "Constructing a consistent and continuous cyanobacteria Bloom monitoring product from multi-mission ocean color instruments," *Remote Sens.*, vol. 15, no. 22, p. 5291, Nov. 2023.

[62] K. Shi et al., "Remote chlorophyll—A estimates for inland waters based on a cluster-based classification," *Sci. Total Environ.*, vol. 444, pp. 1–15, Feb. 2013.

[63] M. E. Smith, L. Robertson Lain, and S. Bernard, "An optimized chlorophyll a switching algorithm for MERIS and OLCI in phytoplankton-dominated waters," *Remote Sens. Environ.*, vol. 215, pp. 217–227, Sep. 2018.

[64] T. Kutser, "Quantitative detection of chlorophyll in cyanobacterial Blooms by satellite remote sensing," *Limnol. Oceanogr.*, vol. 49, no. 6, pp. 2179–2189, Nov. 2004.

[65] L. Qi, C. Hu, P. M. Visser, and R. Ma, "Diurnal changes of cyanobacteria Blooms in Taihu Lake as derived from GOCI observations," *Limnol. Oceanogr.*, vol. 63, no. 4, pp. 1711–1726, Jul. 2018.

[66] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 785–794.

[67] C. Huang et al., "Assessment of NIR-red algorithms for observation of chlorophyll—A in highly turbid inland waters in China," *ISPRS J. Photogramm. Remote Sens.*, vol. 93, pp. 29–39, Jul. 2014.

[68] K. Shi et al., "Long-term MODIS observations of cyanobacterial dynamics in lake Taihu: Responses to nutrient enrichment and meteorological factors," *Sci. Rep.*, vol. 7, no. 1, p. 40326, Jan. 2017.

[69] Z. Cao et al., "MODIS observations reveal decrease in lake suspended particulate matter across China over the past two decades," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113724.

[70] C. Neil, E. Spyrakos, P. D. Hunter, and A. N. Tyler, "A global approach for chlorophyll-a retrieval across optically complex inland waters based on optical water types," *Remote Sens. Environ.*, vol. 229, pp. 159–178, Aug. 2019.

[71] K. H. Hamed and A. Ramachandra Rao, "A modified mann-Kendall trend test for autocorrelated data," *J. Hydrol.*, vol. 204, nos. 1–4, pp. 182–196, Jan. 1998.

[72] P. K. Sen, "Estimates of the regression coefficient based on Kendall's tau," *J. Amer. Stat. Assoc.*, vol. 63, no. 324, pp. 1379–1389, 1968.

[73] R. O. Gilbert, "Statistical methods for environmental pollution monitoring," Pacific Northwest Nat. Lab. (PNNL), Richland, WA, USA, Tech. Rep. PNL-4660, 1987.

[74] K. Havens, H. Paerl, E. Phlips, M. Zhu, J. Beaver, and A. Srifa, "Extreme weather events and climate variability provide a lens to how shallow lakes may respond to climate change," *Water*, vol. 8, no. 6, p. 229, May 2016.

[75] J. R. Beaver et al., "Extreme weather events influence the phytoplankton community structure in a large lowland subtropical lake (Lake Okeechobee, Florida, USA)," *Hydrobiologia*, vol. 709, no. 1, pp. 213–226, Jun. 2013.

[76] G. Ji, K. E. Havens, J. R. Beaver, and T. L. East, "Recovery of plankton from hurricane impacts in a large shallow lake," *Freshwater Biol.*, vol. 63, no. 4, pp. 366–379, Apr. 2018.

[77] F. Mélin, "Uncertainties in ocean colour remote sensing," Int. Ocean Colour Coordinating Group, Dartmouth, NS, Canada, Tech. Rep. 18, 2019.

[78] X. Liu et al., "Retrieval of chlorophyll—A concentration and associated product uncertainty in optically diverse lakes and reservoirs," *Remote Sens. Environ.*, vol. 267, Dec. 2021, Art. no. 112710.

[79] M. Werther et al., "Characterising retrieval uncertainty of chlorophyll—A algorithms in oligotrophic and mesotrophic lakes and reservoirs," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 279–300, Aug. 2022.

[80] T. T. Wynne, R. P. Stumpf, M. C. Tomlinson, and J. Dyble, "Characterizing a cyanobacterial Bloom in western lake Erie using satellite imagery and meteorological data," *Limnology Oceanogr.*, vol. 55, no. 5, pp. 2025–2036, Sep. 2010.

[81] Y. Yao et al., "Detecting cyanobacterial Blooms in the caloosahatchee river and estuary using PlanetScope imagery and deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4202513.

[82] H. W. Paerl and J. Huisman, "Blooms like it hot," *Science*, vol. 320, no. 5872, pp. 57–58, Apr. 2008.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.



**Cheng Xue** received the B.S. degree in physics and the Ph.D. degree in marine detection technology (ocean optics and ocean color remote sensing) from Ocean University of China, Qingdao, China, in 2016 and 2022, respectively.

He is currently a Post-Doctoral Scholar with the Optical Oceanography Laboratory, College of Marine Science, University of South Florida, St. Petersburg, FL, USA. His research interests include in situ observation for water optical properties, ocean color algorithm development, and the application of satellite radiometric data to the aquatic environment.



**Jennifer P. Cannizzaro** received the B.A. degree in biology from Mount Holyoke College, South Hadley, MA, USA, in 1993, and the M.S. degree in marine science from the University of South Florida, St. Petersburg, FL, USA, in 2004.

She is currently a Research Scientist with the Optical Oceanography Laboratory, College of Marine Science, University of South Florida. Her research interests include developing remote sensing algorithms for monitoring harmful algal blooms using satellite ocean color data.



**Chuanmin Hu** received the B.S. degree in physics from the University of Science and Technology of China, Hefei, China, in 1989, and the Ph.D. degree in physics (environmental optics) from the University of Miami, Coral Gables, FL, USA, in 1997.

He is currently a Professor of optical oceanography with the University of South Florida, St. Petersburg, FL, USA, and also directs the Optical Oceanography Laboratory. He uses laboratory, field, and remote sensing techniques to study marine algal blooms (harmful and nonharmful, macroalgae and microalgae), oil spills, coastal and inland water quality, and global changes.

Dr. Hu served as a Topical Editor on ocean optics and ocean color remote sensing of *Applied Optics* from 2009 to 2014. From 2015 to 2017, he served as a Chief Editor of *Remote Sensing of Environment.*



**Brian B. Barnes** received the B.S. degree in zoology and psychology from the University of Florida, Gainesville, FL, USA, in 2004, the M.S. degree in marine science from the College of William and Mary, Virginia Institute of Marine Science, Gloucester Point, VA, USA, in 2009, and the Ph.D. degree in marine science from the University of South Florida, St. Petersburg, FL, USA, in 2013.

He is currently a Research Assistant Professor with the Optical Oceanography Laboratory, College of Marine Science, University of South Florida. His research interests include improving satellite retrievals in optically complex environments, calibration and validation of ocean color satellite data, assessing the continuity of satellites for ocean color research, and floating matter characterization.



**Yuyuan Xie** received the B.S. and Ph.D. degrees in marine science from Xiamen University, Xiamen, China, in 2007 and 2015, respectively.

He is currently a Research Scientist with the Optical Oceanography Laboratory, College of Marine Science, University of South Florida, St. Petersburg, FL, USA. His research interests include the phytoplankton functional types in relation to marine primary production.



**Lin Qi** received the B.S. degree in geography information systems (GIS) from East China University of Technology, Fuzhou, China, in 2008, the M.S. degree in GIS from Nanjing Normal University, Nanjing, China, in 2011, and the Ph.D. degree in remote sensing and GIS from Nanjing Institute of Geography and Limnology of the Chinese Academy of Sciences, Nanjing, in 2015.

She is currently a Research Scientist with the NOAA Ocean Color Science Team. Her research interests include optical characterization and remote sensing of floating macroalgae, phytoplankton, and organic matter in global oceans and lakes, and their responses to climate variability and human activities.



**Cassondra Armstrong** received the B.S. degree in environmental science from Mary Washington College, Fredericksburg, VA, USA, in 1992, the M.S. degree in biology from East Carolina University, Greenville, NC, USA, in 1998, and the Ph.D. degree in environmental sciences from the University of Virginia, Charlottesville, VA, USA, in 2004.

She is currently a Section Administrator for the Coastal Ecosystem Section in the Applied Sciences Bureau of the South Florida Water Management District, West Palm Beach, FL, USA. Her research interests include water quality, treatment of wetlands, and climate resilience. As part of her water quality responsibilities, understanding harmful algal bloom formation, detection, and management is a priority.



**Zhiqiang Chen** received the B.S. degree in marine science from Xiamen University, Xiamen, China, in 1991, and the Ph.D. degree in marine science from the University of South Florida, St. Petersburg, FL, USA, in 2006.

He is currently a Senior Scientist with South Florida Water Management District, West Palm Beach, FL, USA. His research interests include water quality monitoring with ocean color satellite data and ecological modeling in coastal and estuarine waters.



**Paul R. Jones** received the B.S. degree in biochemistry and molecular biology from the University of Leeds, Leeds, U.K., in 1999, the M.S. degree in marine biology and biochemistry from the College of Earth, Ocean and Environment, University of Delaware, Lewes, Delaware, in 2005, and the Ph.D. degree in marine biology from the Rosenstiel School of Marine and Atmospheric Science, University of Miami, Miami, FL, USA, in 2014.

He was then a National Research Council Post-Doctoral Fellow at NOAA's Atlantic Oceanographic and Meteorological Laboratory, Miami. He currently works as a Lead Scientist with the Lake and River Ecosystems Section of the Applied Sciences Bureau, South Florida Water Management District, West Palm Beach, FL, USA. He has been with SFWMD for seven years, focusing predominantly on water quality in Lake Okeechobee and the surrounding watershed.

Dr. Jones is a member of the Florida Lake Management Society.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:04:52 UTC from IEEE Xplore. Restrictions apply.